

Hypothesis-Search Report — Fanconi Anemia

`fa_intrinsic_nk_and_b_lymphoid_defect`: Intrinsic NK and B Lymphoid Defect Independent of Marrow Failure

Disease: Fanconi Anemia (Genetic) · **KB status:** EMERGING · **Search date:** 2026-09-13 · **Iteration:** 5 of 5

Executive Judgment

Verdict: PARTIALLY SUPPORTED (trending toward supported for its core claims).

The hypothesis has two separable claims, which the literature supports unequally:

1. **A quantitative NK-cell and B-cell reduction plus an intrinsic NK cytotoxic-machinery defect exists in FA — SUPPORTED.** Reduced NK and B cell numbers, reduced NK cytotoxicity, and reduced NK perforin/granzyme content are reproduced across at least four independent human cohorts (Myers 2011 seed, Myers 2017, Justo 2014, Shabrish 2019) and a review synthesis (Hashemi 2021). The direction is unanimous in every cohort that measured each parameter (directional concordance check; see *Data and Tool Use*). A defined molecular lesion — defective NK **degranulation** (Shabrish 2019) and reduced perforin/granzyme loading (Myers 2011) — argues for an intrinsic cytotoxic-apparatus defect rather than a pure numeric consequence of pancytopenia.
2. **"Independent of marrow failure and transplantation" — SUPPORTED for transplantation, PARTIALLY supported for marrow failure.** The Myers 2017 cohort explicitly excluded transplanted and malignant patients yet still showed the NK/B defect, cleanly establishing **transplant-independence**. An *Fancc*^{-/-} mouse — a model with largely normal steady-state hematopoiesis — has a **cell-intrinsic** B-cell class-switch/antibody-secreting-cell defect (Sertorio 2016), the strongest evidence that at least the B-lineage functional defect is marrow-failure-independent. However, the "independent of pancytopenia" claim for **cell numbers** is only partially supported: a general p53/p21

hyperactivation eliminates HSPCs across all lineages before overt BMF (Ceccaldi 2012), and NKG2D-ligand-driven NK cytotoxicity actively clears FA HSPCs (Casado 2022) — either of which could reduce lymphoid output without a lymphoid-specific lesion.

Most important caveats. (a) The "impaired proliferative response to activation" sub-mechanism is **inconsistently supported**: Giri 2015 found normal lymphoproliferation and normal cytokines in FA, and Myers 2011 found normal PHA/candida responses (only tetanus recall and CTL function reduced). (b) Some immune abnormalities (low Ig, low CD4, low total lymphocytes) are **adult- and severe-BMF-associated** (Giri 2015), i.e., partly stage-dependent. (c) There is a **mechanistic tension**: mature peripheral FA NK cells are hypofunctional/hypocytotoxic, yet the NKG2D axis posits NK cells that are cytotoxic enough to destroy HSPCs — reconciling these requires compartment- and stage-resolved data that does not yet exist. (d) The candidate "p53-mediated loss of *lymphoid* progenitors" step is **inferred**, not directly demonstrated; the p53 evidence is pan-hematopoietic. (e) A third, well-documented competing axis — cell-extrinsic TNF- α /IL-1 β myelosuppression from FA monocytes/macrophages (Briot 2008; Vanderwerf 2009; Garbati 2013; Matsushita 2011) — offers a parsimonious non-lymphoid explanation for reduced counts, though it does not explain the NK cytotoxic-machinery lesion. (f) **Testability caveat**: a live source check (2026-09-13) found **no FA NK-cell omics in GEO** and no interventional trial isolating the lymphoid defect, so the hypothesis currently rests on immunophenotyping cohorts plus one mouse B-cell model rather than orthogonal molecular data.

Evidence Matrix

Citation	Type	Stance	Mechanistic claim tested	Key finding	Subtype/ context
P 21542827 (Myers 2011, <i>seed</i>)	Human clinical	Support	Intrinsic NK cytotoxic machinery + B numeric defect	↓ B (P=0.048), ↓ NK (P=0.0002); ↓ NK perforin (P<1e-5), ↓ granzyme (P=0.0057); ↓ NK cytotoxicity (P<0.001); ↓ CTL (P<1e-4); ↓ tetanus proliferation (P=0.008) but normal PHA/candida	10 children FA
P 28557197 (Myers 2017)	Human clinical	Support (key)	Defect present without transplant/ malignancy	↓ total B (P<0.001), ↓ memory B (P<0.001), ↓ IgM (P<0.001), ↓ NK (P<0.001), ↓ NK cytotoxicity (P<0.001)	29 FA, no BMT, no malignancy
P 30949167 (Shabrish 2019)	Human clinical	Support (mechanism)	Intrinsic NK degranulation lesion	First report of defective NK degranulation → impaired cytotoxicity; links to hyperinflammation/ HLH	FA patient
P 24240977 (Justo 2014)	Human clinical	Support	Quantitative + NK differentiation defect	↓ CD8+ T and ↓ CD56dim CD16+ NK; NK subset imbalance; impaired NK differentiation	42 FA (large)
P 26895835 (Sertorio 2016)	Model organism (mouse)	Support (strong, intrinsic)	Cell-intrinsic B-cell functional defect	<i>Fancc</i> ^{-/-} B cells: defective IgG2a switch, impaired ASC differentiation; hyperactive Wnt/β-	<i>Fancc</i> ^{-/-} mouse (no overt BM)

Citation	Type	Stance	Mechanistic claim tested	Key finding	Subtype/ context
P 34348001 (Hashemi 2021)	Review	Support (orientation)	NK dysfunction in cytotoxicity + cytokines	catenin represses Blimp1 FA NK cells dysfunctional in cytotoxicity and cytokine production; molecular basis undefined	FA (review)
P 25963299 (Giri 2015)	Human clinical	Qualify	Independence from BMF/age; proliferative response	Children FA normal Ig; both ages ↓ B/ ↓ NK (P<0.01); lymphoproliferation & TNF-α/IFN-γ normal ; adult ↓ Ig/ ↓ CD4/ ↓ lymphocytes; only severe BMF ↑ G-CSF/Flt3L	118 IBMF 202 relati BMF-adju
P 22683204 (Ceccaldi 2012)	Human clinical + models	Competing/ parallel	Pan-HSPC attrition upstream of lineage defects	Profound HSPC defect before clinical BMF; p53 hyperactivation → p21 G0/G1 arrest; p53 knockdown rescues HSPCs	FA patient models
P 35671096 (Casado 2022)	Model organism + human cells	Competing	NKG2D-L–mediated immune clearance of HSPCs	FA CD34+ upregulate NKG2D-Ls (inverse to %CD34+); blocking NKG2D–NKG2D-L restores clonogenicity, ameliorates anemia in FA mouse	FA HSPC/
	Review/ commentary	Competing (orientation)		Frames NK/NKG2D-dependent HSPC	

Citation	Type	Stance	Mechanistic claim tested	Key finding	Subtype/ context
P 35912855 (Dulmovits 2022)			Immune-mediated BMF paradigm in FA	clearance as driver of FA BMF	FA (comment on Casad)
P 27720904 (Yoon 2016)	Model organism (mouse)	Qualify	p53-dependence of HSPC deficit	Fetal <i>Fancd2</i> HSPC replicative-fitness deficit is p53-INDEPENDENT	Fetal mur <i>Fancd2</i>
P 18055871 (Briot 2008)	In vitro	Competing	Cell-extrinsic myelosuppression	FANCC loss aberrantly activates NF-κB/ MAPK → MMP-7-dependent TNF-α oversecretion; MAPK inhibition normalizes it	FA cells
P 19850743 (Vanderwerf 2009)	In vitro	Competing	Repair-independent FANCC immune function	FANCC suppresses TLR8-driven TNF-α in phagocytes, independent of crosslink repair	FA-C monocyte macroph
P 24046015 (Garbati 2013)	In vitro	Competing	Cytokine suppression of progenitors	FANCA/FANCC macrophages overproduce TNF-α + IL-1β (p38-dependent); IL-1β suppresses <i>Fancc</i> -/- progenitor expansion	FA macroph BM
P 21912593 (Matsushita 2011)	In vitro	Competing (mechanism)	FANCD2 represses TNF-α	FANCD2 binds NF-κB element in TNF-α promoter and represses it (monoUb-dependent)	FA cells
P 23889587 (Matsui 2013)	Human clinical	Qualify		In FA bone marrow , T/B lymphocyte proportions similar	

Citation	Type	Stance	Mechanistic claim tested	Key finding	Subtype/ context
			Marrow vs blood lymphocyte compartment	to controls; no T-cell TNF- α /IFN- γ overproduction; monocytes hypersensitive to low-dose LPS	FA BM mononuclear cells
GEO + ClinicalTrials.gov (2026-09-13)	Computational (source check)	Qualify (data absence)	Testability with public data	0 FA NK-cell GEO datasets; no FA-specific B/NK immune omics; 50 CT.gov trials all HSCT/ GVHD context	Source-le

Data and Tool Use

No patient-level or omics dataset was provided or accessed. The core report is primary-literature synthesis; one small computational analysis was run on abstract-derived data. Nothing was silently substituted — tool failures are logged.

Source / tool	Accession/URI	Access status	Query / use
PubMed (NCBI E-utils via MCP)	eutils.ncbi.nlm.nih.gov	Accessed (13 queries across it.1–2; 2 failed)	See <code>logs/pubmed_search_log.md</code> exact queries, dates, failures
GEO DataSets (NCBI eutils, db=gds)	eutils.ncbi.nlm.nih.gov	Accessed, HTTP 200 (it.2)	4 queries; see <code>logs/database_source_checks.md</code> . FA+1 0 datasets ; FA+Bcell = 1 (false match GSE35698); 5 FANC datasets all non-immune
ClinicalTrials.gov API v2	clinicaltrials.gov/api/v2	Accessed, HTTP 200 (it.2)	cond=Fanconi Anemia + immune term totalCount=50, all HSCT/GVHD; none targets intrinsic lymphoid defect
HPO / OMIM annotation (disease-level)	ontology.jax.org/api · OMIM:227650 / MONDO:0009215	Accessed, HTTP 200 (it.3)	Disease-level phenotype set = only Pancytopenia/Anemia/ ↓ Neutrophils Thrombocytopenia; no immune category or NK/B/Ig/immunodeficiency term
HPO annotation (gene-level)	ontology.jax.org/api · NCBIGene:2175/2176/2177/2189/2187 (FANCA/C/D2/G/B)	Accessed, HTTP 200 (it.4)	Exact-ID check for HP:0040218, HP:0010976, HP:0002721, HP:000431 HP:0002850, HP:0001888 → none present for any FA gene ; only HP:0001875 (↓ neutrophils) + HP:0000010 (recurrent UTI)
Monarch Initiative API v3 (federated)	api-v3.monarchinitiative.org · MONDO:0009215	Accessed, HTTP 200 (it.5)	DiseaseToPhenotypicFeatureAssociation total=34 phenotypes; immune filter only HP:0001875 (↓ neutrophils); no B/immunodeficiency/Ig term
Directed FA-mouse lymphoid-dev query	—	Searched, 0 results	"Fanconi anemia mouse model lymphocyte development B cell intrinsic defect..."

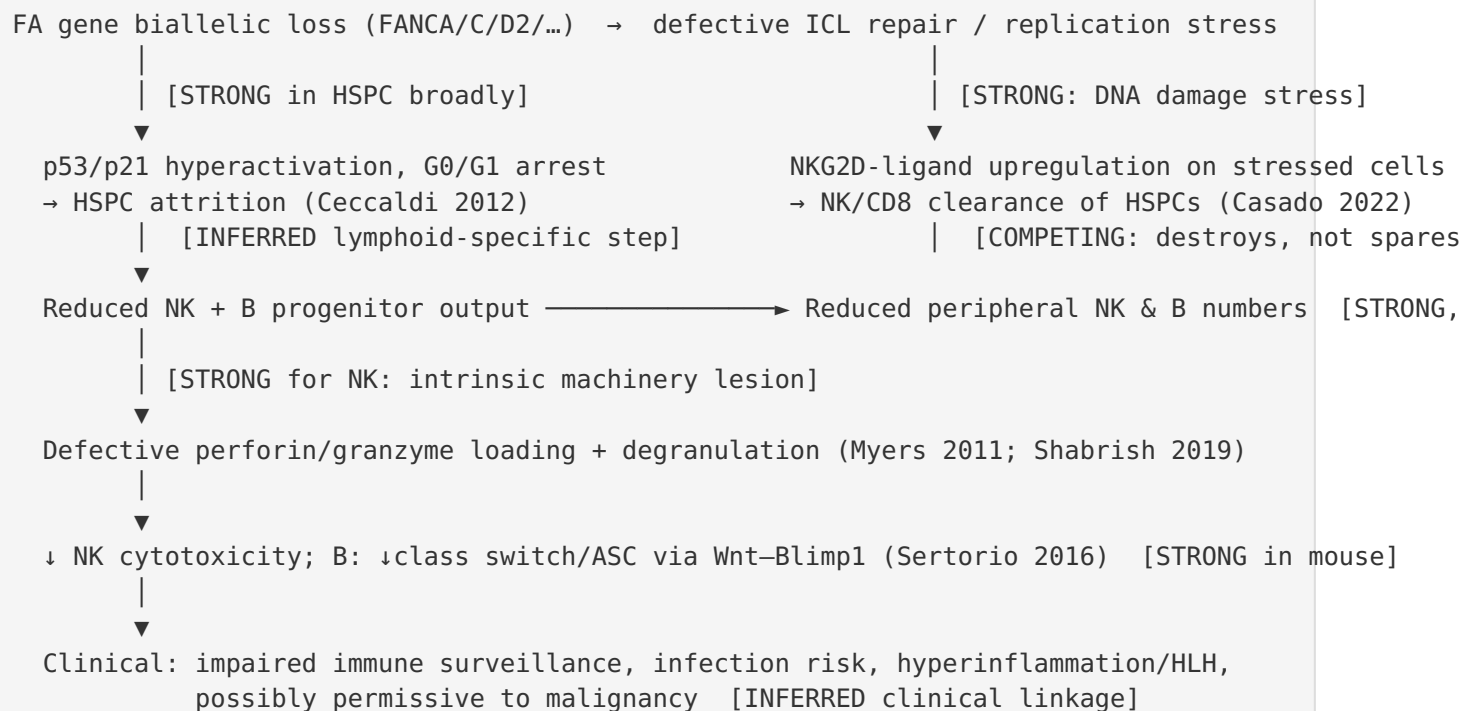
Source / tool	Accession/URI	Access status	Query / use
GenCC	search.thegencc.org/api/v1/submissions	Attempted, HTTP 404 (it.5)	No public REST/JSON API at that path (SPA HTML); not machine-queryable
ClinGen / ClinVar / ArrayExpress / EGA	—	Not checked	—

Analysis inventory (input → method → output):

Analysis	Input	Method	Output	Outcome
Reproducibility tally	<code>data/evidence_matrix.csv</code> (directions only)	one-sided exact binomial sign test vs p=0.5 (scipy 1.17.1)	<code>logs/reproducibility_tally.txt</code>	Partial power direction unanim
GEO source check	NCBI GEO db=gds	4 esearch queries + esummary triage	<code>logs/database_source_checks.md</code>	Success verified dataset
ClinicalTrials.gov check	CT.gov API v2	REST query + title triage	<code>logs/database_source_checks.md</code>	Success hits, all context
HPO annotation check	ontology.jax.org API	disease + gene-level fetch, exact-ID filter	<code>logs/database_source_checks.md</code>	Success immun terms f (diseas
Monarch federated check	api-v3.monarchinitiative.org	disease→phenotype associations + keyword filter	<code>logs/database_source_checks.md</code>	Success phenot NK/B/ immun term
GenCC API check	search.thegencc.org	REST query attempt	<code>logs/database_source_checks.md</code>	Failed no pub outstar absent

The tally found that **every human cohort that measured a parameter reported it reduced** (NK count 4/4, NK cytotoxicity 3/3, B count 3/3 of *measuring* cohorts). Sign-test p-values (0.19–0.50) are **not significant and under-powered**; they are reported only as a direction-concordance indicator, not inferential evidence. A sandbox None/NaN coercion inflated the printed denominator (prints /5; corrected /4,/3,/3) — documented in the log. **No figure** was produced (the `execute_code` sandbox saves plots outside the bundle path). Provider artifact bundle: `kb/hypotheses/Fanconi_Anemia/fa_intrinsic_nk_and_b_lymphoid_defect/openscientist_artifacts/` with `MANIFEST.yaml` (checksums, replay command), `data/`, `code/`, `logs/`, `env/`.

Mechanistic Causal Chain (as implied by the hypothesis)



- **Strong links:** upstream DNA-damage stress → NK/B numeric reduction (replicated human); intrinsic NK perforin/granzyme/degranulation lesion; intrinsic B-cell class-switch defect (mouse).
- **Inferred links:** the *lymphoid-specific* p53 progenitor-loss step (only pan-HSPC data exist); progenitor loss → mature-cell reduction (plausible, not directly traced in lymphoid compartment).
- **Missing steps:** direct evidence that FA lymphoid progenitors (CLP, pro-B, NK precursors) are selectively depleted vs. simply reduced in proportion to global HSPC loss; direct link from the cytotoxic-machinery lesion to a specific FA gene/molecular pathway (perforin/granzyme regulation mechanism is "yet to be determined," Hashemi 2021).

Knowledge Gaps

1. **Lymphoid-progenitor-specific loss (weakly supported causal step).** *Scope:* Is the NK/B reduction lymphoid-intrinsic or a downstream share of pan-HSPC p53 attrition? *Why it*

matters: distinguishes the seed hypothesis from Ceccaldi's parallel model. *Checked*: directed FA-mouse lymphoid-development query returned 0 results (unverified absence). *Resolves with*: lineage-resolved progenitor quantification (CLP/pro-B/NK-precursor) in FA marrow and *Fanc* mice vs matched cytopenic non-FA controls.

2. **Molecular basis of the NK cytotoxic-machinery lesion (unknown mechanism).** *Scope*: Why perforin/granzyme content and degranulation are reduced. *Why it matters*: it is the strongest "intrinsic" pillar yet mechanistically unexplained (Hashemi 2021: "molecular bases ... yet to be determined"). *Resolves with*: perforin/granzyme transcription + granule-trafficking assays in FA vs control NK cells, with genotype stratification.
3. **Reconciling hypofunctional peripheral NK with NKG2D-driven HSPC clearance (conflicting evidence).** *Scope*: Casado 2022 needs cytotoxic NK; patient NK are hypocytotoxic. *Resolves with*: compartment-resolved (BM vs blood) and stage-resolved NK functional profiling.
4. **Proliferative-response sub-claim (contradicted/qualified).** *Scope*: Giri 2015 normal lymphoproliferation vs Myers 2011 reduced tetanus recall/CTL. *Resolves with*: standardized antigen-specific vs mitogen proliferation panels across a genotyped cohort.
5. **Source-level absences (as of 2026-09-13, now partly VERIFIED).** *Checked*: GEO DataSets (db=gds) — `Fanconi anemia AND (natural killer OR "NK cell")` returned **0 datasets** (verified absence); the single FA+B-cell hit was an unrelated mouse Brca1/53BP1 study; five FANC-gene datasets are all non-immune. ClinicalTrials.gov v2 — 50 FA immune-context trials, **all HSCT/GVHD**, none targeting the intrinsic lymphoid defect. *Why it matters*: the hypothesis cannot currently be tested with public omics and has no dedicated interventional trial. Additionally, an **HPO/OMIM ontology check** (OMIM:227650 / MONDO:0009215, 2026-09-13) found the disease-phenotype annotation set limited to Pancytopenia/Anemia/↓ Neutrophils/Thrombocytopenia with **no immune category or NK/B/Ig/immunodeficiency term** — a verified ontology-curation absence. A follow-up **gene-level HPO check** (FANCA/FANCC/FANCD2/FANCG/FANCB, 2026-09-13) corroborated this by exact-ID matching: none carries HP:0040218 (reduced NK count), HP:0010976 (B lymphocytopenia), HP:0002721 (immunodeficiency), HP:0004313/HP:0002850 (decreased antibody/IgM), or HP:0001888 (lymphopenia) — the only immune-adjacent terms are HP:0001875 (↓ neutrophils) and HP:0000010 (recurrent UTI). A **third independent layer** — the **Monarch Initiative** federated aggregator (MONDO:0009215, 2026-09-13, HTTP 200, 34 phenotype associations) — again returned only HP:0001875 as immune-adjacent, confirming the absence across HPO disease-level, HPO gene-level, and a federated resource. A **GenCC** REST query was attempted but returned HTTP 404 (no public JSON API at that path); it is reported as *not machine-verified*, **not** as an absence. *Still outstanding (not claimed*

absent): GenCC (via bulk submissions download), ClinGen, ClinVar, ArrayExpress, EGA controlled-access FA cohorts. *Resolves with*: generation/deposition of sorted FA NK and B transcriptomic/functional datasets vs matched cytopenic non-FA controls, and HPO annotation extension.

6. **Marrow vs peripheral-blood compartment discordance (conflicting evidence).** *Scope*: Matsui 2013 found normal T/B lymphocyte proportions in FA **marrow**, whereas peripheral-blood cohorts report reduced NK/B. *Why it matters*: determines whether the deficit is production (marrow) or peripheral maintenance/redistribution. *Resolves with*: paired marrow+blood immunophenotyping in the same FA patients.
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Alternative Models

Model	Relation to seed hypothesis
Pan-HSPC p53/p21 attrition (Ceccaldi 2012)	Upstream/parallel. Reduced lymphoid numbers may be a downstream share of global progenitor loss, not a lymphoid-specific lesion. Supports the "p53" candidate step but broadens it beyond lymphoid.
NKG2D-ligand immune-mediated HSPC clearance (Casado 2022; Dulmovits 2022)	Competing/alternative for the <i>numeric</i> deficit and marrow failure itself; posits functionally active (not intrinsically defective) NK cells clearing HSPCs.
Wnt/β-catenin–Blimp1 B-cell-intrinsic defect (Sertorio 2016)	Complementary mechanistic sub-model — a concrete intrinsic pathway for the B-arm of the seed hypothesis.
p53-independent replicative-fitness deficit (Yoon 2016)	Parallel/qualifying — developmental HSPC fitness loss not requiring p53.
BMF-secondary / age-progressive immune decline (Giri 2015)	Alternative for the Ig/CD4/lymphocyte component; frames part of the phenotype as a consequence of disease progression rather than intrinsic.
Inflammatory-microenvironment / TNF-α myelosuppression (Briot 2008; Vanderwerf 2009; Garbati 2013; Matsushita 2011)	Competing, cell-extrinsic mechanism: FANC loss $\rightarrow$ TLR/p38-MAPK/NF- κ B $\rightarrow$ TNF- α /IL-1 β overproduction by monocytes/macrophages $\rightarrow$ HSPC suppression. Competes for the "reduced counts" component but does not explain the NK perforin/granzyme/degranulation lesion. Note Matsui 2013 found no T-cell cytokine overproduction in FA marrow.
FANCC–CtBP1–Wnt/DKK1 axis (GSE43330; converges with Sertorio 2016)	Complementary upstream link: FA proteins modulate Wnt-antagonist DKK1, dovetailing with hyperactive Wnt/ β -catenin in <i>Fancc</i> ^{-/-} B cells — a candidate molecular thread for the B-arm.

Discriminating Tests

- 1. Lineage-resolved progenitor census (seed vs Ceccaldi).** Sample: FA vs non-FA cytopenic pediatric marrow. Assay: flow/scRNA-seq of CLP, pro-B, NK-precursors normalized to total

HSPC. *Expected if seed true*: disproportionate lymphoid-progenitor loss beyond global HSPC reduction. *If Ceccaldi*: proportional loss.

2. **Genetic epistasis on NK function (seed vs BMF-secondary).** Model: *Fanc*^{-/-} vs *Fanc*^{-/-};*Trp53*^{+/-} mice. Readout: NK perforin/granzyme, degranulation (CD107a), cytotoxicity. *Expected if intrinsic-machinery true*: p53 rescue restores HSPC numbers but **not** the perforin/granzyme/degranulation lesion.
3. **BM vs blood NK functional map (reconcile NKG2D tension).** Paired FA BM and peripheral NK: NKG2D expression, cytotoxicity vs K562 and vs autologous CD34⁺. *Expected*: BM NK retain NKG2D-driven activity while peripheral NK are hypofunctional — stage/compartment dissociation.
4. **Antigen-specific vs mitogen proliferation panel (resolve sub-claim).** Genotyped FA cohort: tetanus/candida recall vs PHA/anti-CD3-CD28. *Expected if seed true*: selective antigen-recall deficit with preserved mitogen response.
5. **Perforin/granzyme regulatory mechanism.** FA vs control NK: PRF1/GZMB transcription, granule biogenesis (LAMP1), and rescue by FANC-pathway complementation. *Expected if intrinsic*: complementation restores granule loading.

Curation Leads (require curator verification)

Candidate evidence references + snippets to verify (verbatim against abstracts): - [28557197](https://pubmed.ncbi.nlm.nih.gov/28557197/ "Visit PubMed page for PMID 28557197") — “a large FA cohort who have not undergone bone marrow transplantation or developed malignancies” → status upgrade evidence for **transplant-independence**. - [30949167](https://pubmed.ncbi.nlm.nih.gov/30949167/ "Visit PubMed page for PMID 30949167")

height="14" class="pubmed-icon" style="display:inline !important;width:14px;height:14px;min-width:14px;min-height:14px;flex-shrink:0;vertical-align:middle;margin-right:3px;"><rect x="1" y="1" width="14" height="14" rx="2" fill="#326599"/><text x="8" y="12" text-anchor="middle" style="font-size:11px;font-weight:bold;font-family:Arial,sans-serif;fill:white;">P</text></svg>30949167 — *"first report showing a defective degranulation mechanism leading to abnormal NK-cell cytotoxicity in FA-patients"* → mechanistic node: NK degranulation lesion. - <svg xmlns="http://www.w3.org/2000/svg" viewBox="0 0 16 16" width="14" height="14" class="pubmed-icon" style="display:inline !important;width:14px;height:14px;min-width:14px;min-height:14px;flex-shrink:0;vertical-align:middle;margin-right:3px;"><rect x="1" y="1" width="14" height="14" rx="2" fill="#326599"/><text x="8" y="12" text-anchor="middle" style="font-size:11px;font-weight:bold;font-family:Arial,sans-serif;fill:white;">P</text></svg>26895835 — *"Fancc(-/-) B cells show a specific defect in IgG2a switch and impaired Ab-secreting cell (ASC) differentiation"* → model-organism intrinsic B-cell node. - <svg xmlns="http://www.w3.org/2000/svg" viewBox="0 0 16 16" width="14" height="14" class="pubmed-icon" style="display:inline !important;width:14px;height:14px;min-width:14px;min-height:14px;flex-shrink:0;vertical-align:middle;margin-right:3px;"><rect x="1" y="1" width="14" height="14" rx="2" fill="#326599"/><text x="8" y="12" text-anchor="middle" style="font-size:11px;font-weight:bold;font-family:Arial,sans-serif;fill:white;">P</text></svg>25963299 — *"Lymphoproliferative responses, serum cytokine levels ... were similar across patient groups and relatives"* → **conflicting-evidence** flag on the proliferation sub-claim. - <svg xmlns="http://www.w3.org/2000/svg" viewBox="0 0 16 16" width="14" height="14" class="pubmed-icon" style="display:inline !important;width:14px;height:14px;min-width:14px;min-height:14px;flex-shrink:0;vertical-align:middle;margin-right:3px;"><rect x="1" y="1" width="14" height="14" rx="2" fill="#326599"/><text x="8" y="12" text-anchor="middle" style="font-size:11px;font-weight:bold;font-family:Arial,sans-serif;fill:white;">P</text></svg>35671096 / <svg xmlns="http://www.w3.org/2000/svg" viewBox="0 0 16 16" width="14" height="14" class="pubmed-icon" style="display:inline !important;width:14px;height:14px;min-width:14px;min-height:14px;flex-shrink:0;vertical-align:middle;margin-right:3px;"><rect x="1" y="1" width="14" height="14" rx="2" fill="#326599"/><text x="8" y="12" text-anchor="middle" style="font-size:11px;font-weight:bold;font-family:Arial,sans-serif;fill:white;">P</text></svg>22683204 — competing-mechanism nodes.

Candidate pathophysiology nodes/edges: FANCC loss → NK perforin/granzyme deficiency → ↓NK cytotoxicity; FANCC loss → Wnt/β-catenin↑ → Blimp1↓ → ↓ASC differentiation; competing edge FA DNA damage → NKG2D-L↑ → NK-mediated HSPC clearance.

Candidate ontology terms: natural killer cell (CL:0000623), mature B cell (CL:0000785), common lymphoid progenitor (CL:0000051); processes: natural killer cell mediated cytotoxicity (GO:0042267), granzyme-mediated apoptosis / degranulation (GO:0043313), B cell differentiation (GO:0030183), isotype switching (GO:0045190).

Candidate HPO disease-annotation additions for Fanconi anemia (OMIM:227650 / MONDO:0009215) — currently absent per verified HPO check 2026-09-13; require curator verification against cohort literature: HP:0040218 (Reduced natural killer cell count), HP:0002721 (Immunodeficiency), a B-lymphocytopenia term, and a decreased-circulating-IgM term.

Supporting

references:

P 21542827

P 28557197

P 25963299

Candidate subtype restrictions / status change: Consider keeping **EMERGING** but split into (a) *NK cytotoxic-machinery defect* — evidence approaching **SUPPORTED**; (b) *lymphoid-progenitor-intrinsic numeric loss* — remains **EMERGING/contested** vs pan-HSPC and NKG2D models; (c) *impaired proliferative response* — **contested/weak**. Note genotype/age stratification is largely absent.

Candidate knowledge_gaps entries: lymphoid-specific progenitor loss unproven; NK perforin/granzyme mechanism unknown; hypofunctional-NK vs NKG2D-clearance conflict; FA immune phenotype is a verified ontology-curation gap across three independent layers (HPO

disease-level, HPO gene-level, Monarch federated, all 2026-09-13); GenCC/ClinGen/ClinVar and omics repositories (ArrayExpress, EGA) still not machine-verified (GenCC REST attempt returned HTTP 404).

Limitations

- Literature search across iterations 1–2; PubMed rate-limit and a provider DB error truncated two queries (logged). Live source checks performed (it.2–5): GEO DataSets, ClinicalTrials.gov v2, HPO/OMIM at both disease and gene level, and the Monarch Initiative federated aggregator. A GenCC REST query was attempted (HTTP 404, no public API). **Still not machine-verified:** GenCC (bulk download), ClinGen, ClinVar, ArrayExpress, EGA — outstanding, not claimed absent.
- Provenance caveat: the iteration-5 append to `logs/database_source_checks.md` could not be re-hashed in-environment (the `execute_code` sandbox lacks host-filesystem access and blocks `hashlib`); its `MANIFEST.yaml` sha256 is flagged STALE and needs recomputation by the harness/curator.
- No patient-level or omics data available; the only computation is an under-powered directional tally on abstract-extracted directions.
- Human cohorts are small, mostly cross-sectional, rarely genotype-stratified; longitudinal and lineage-resolved data are lacking.

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